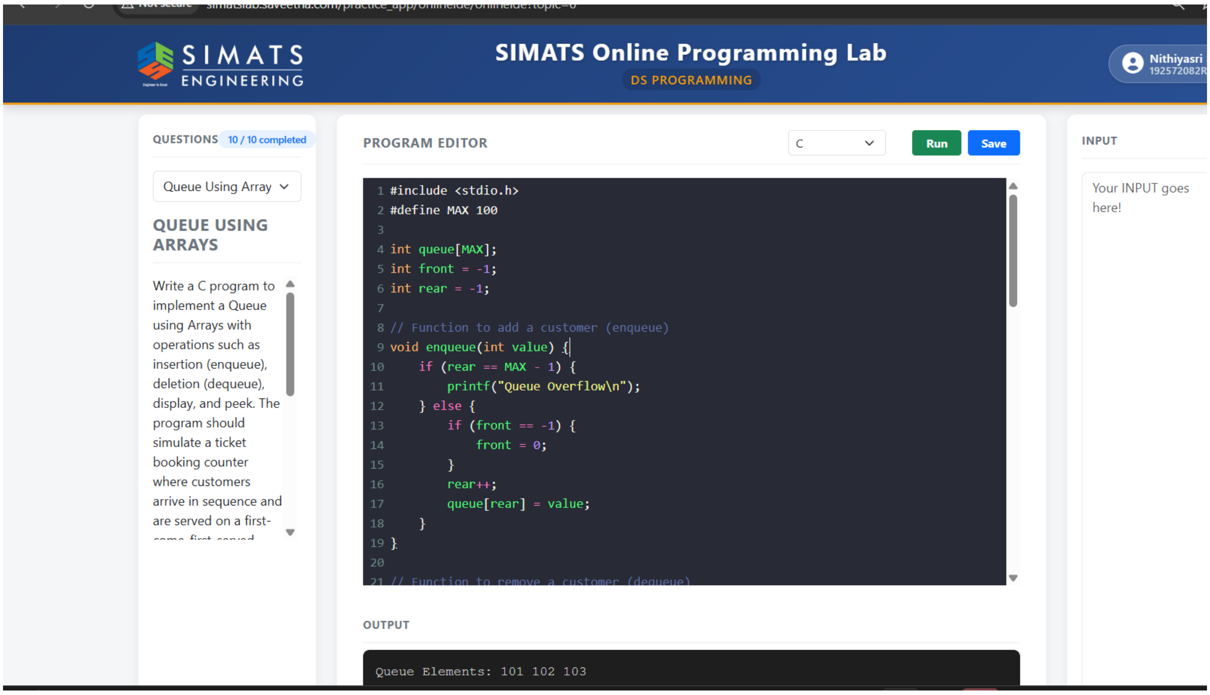
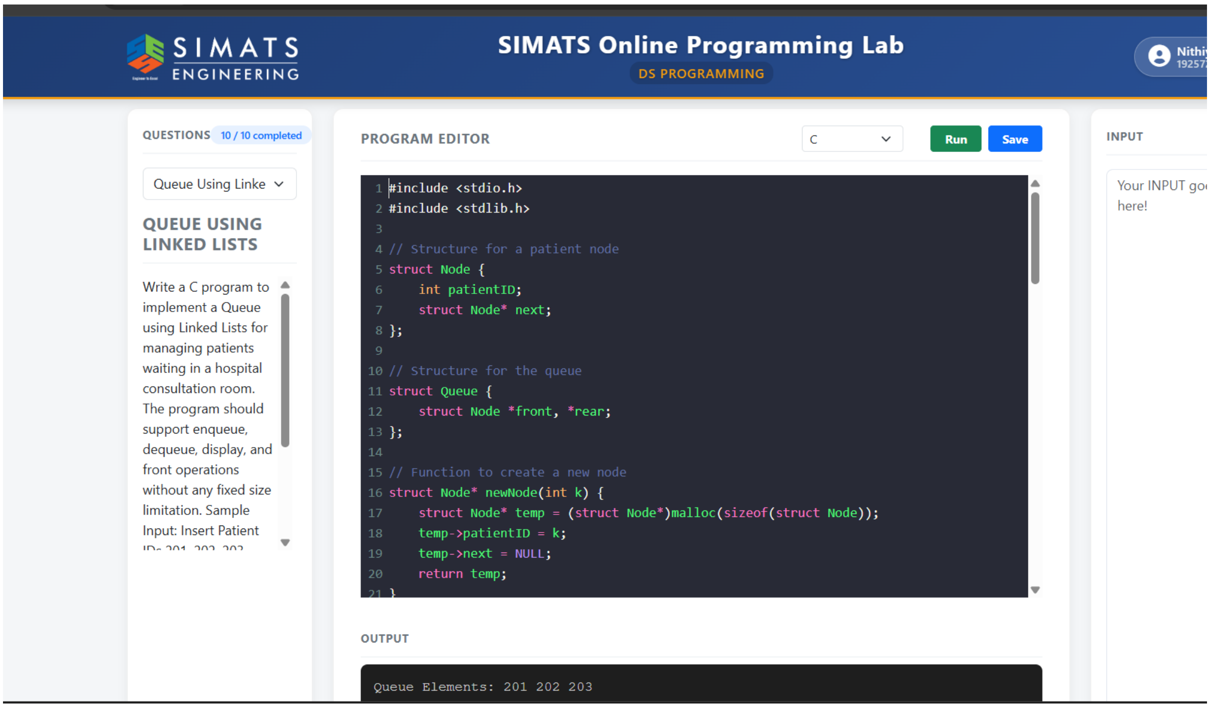
LAB PROGRAMS

QUEUE





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QUESTIONS 10 / 10 completed

Double Ended Que

DOUBLE ENDED QUEUE

Write a C program to implement a Double Ended Queue (Deque) using Arrays where insertion and deletion can be performed from both front and rear ends. Consider a warehouse inventory management system where goods can be added or removed from either side based

PROGRAM EDITOR

C

Run Save

INP

```c
#include <stdio.h>
#define MAX 10

int deque[MAX];
int front = -1;
int rear = -1;

// Function to check if the deque is full
int isFull() {
    return ((front == 0 && rear == MAX - 1) || (front == rear + 1));
}

// Function to check if the deque is empty
int isEmpty() {
    return (front == -1);
}

// Insert at Front
void insertFront(int val) {
    if (isFull()) {
        printf("Queue Overflow\n");
```

OUTPUT

```
Deleted Element: 200
```

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QUESTIONS 10 / 10 completed

Priority Queue Usir

PRIORITY QUEUE USING ARRAYS

Write a C program to implement a Priority Queue using Arrays for processing emergency service requests. Requests with higher priority must be processed before lower-priority requests irrespective of arrival order. Sample Input: Insert (101, Priority 3), Insert

PROGRAM EDITOR

C

Run Save

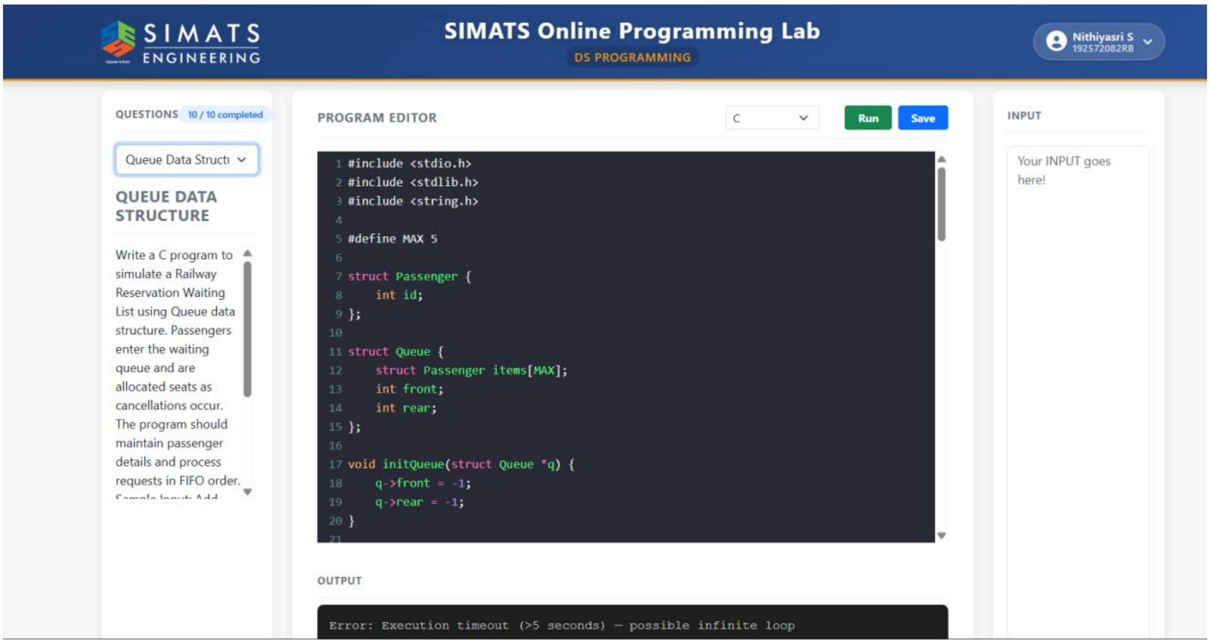
INPUT

Your INPUT here!

```c
#include <stdio.h>
#define MAX 100

typedef struct {
    int id;
    int priority;
} Request;

Request pq[MAX];
int size = 0;

void insert(int id, int priority) {
    if (size >= MAX) {
        printf("Queue Overflow!\n");
        return;
    }

    pq[size].id = id;
    pq[size].priority = priority;
    size++;
```

OUTPUT

```
Processed Request: 102
```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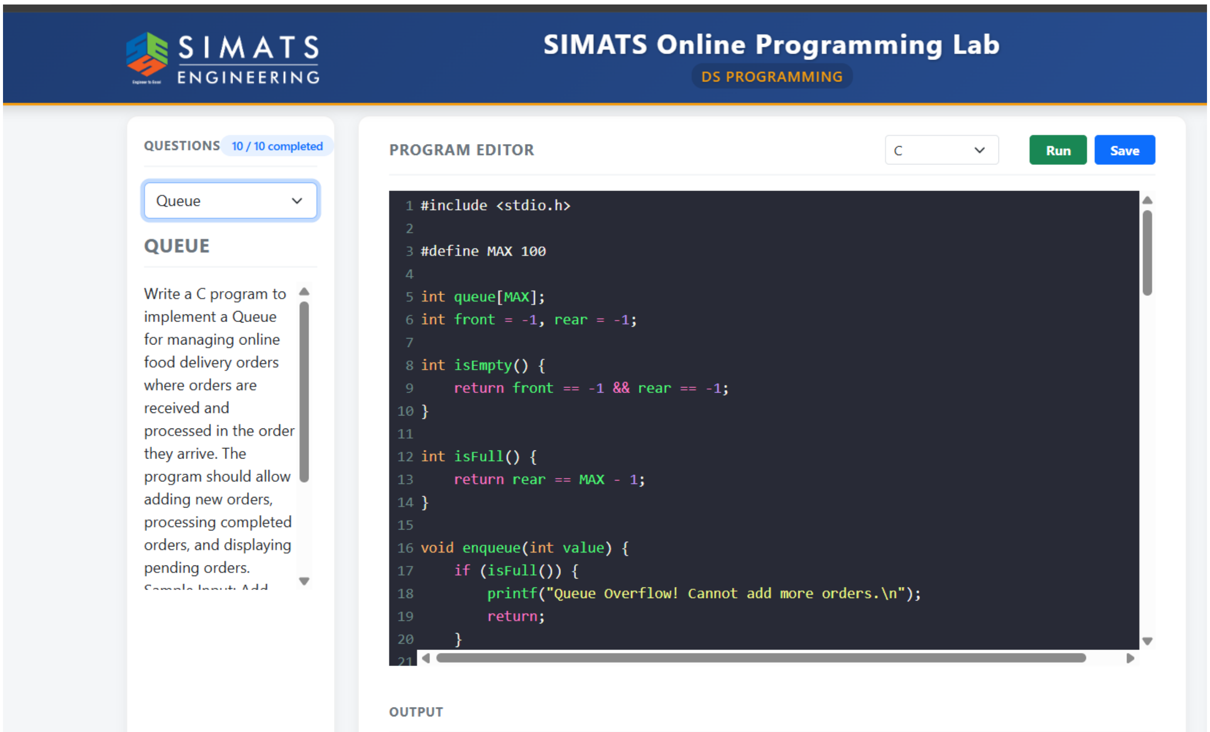
QUESTIONS 10 / 10 completed
Queue Data Structu
QUEUE DATA STRUCTURE
Write a C program to simulate a Railway Reservation Waiting List using Queue data structure. Passengers enter the waiting queue and are allocated seats as cancellations occur. The program should maintain passenger details and process requests in FIFO order.
PROGRAM EDITOR
C
Run
Save
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#define MAX 5
struct Passenger {
int id;
};
struct Queue {
struct Passenger items[MAX];
int front;
int rear;
};
void initQueue(struct Queue *q) {
q->front = -1;
q->rear = -1;
}
OUTPUT
Error: Execution timeout (>5 seconds) — possible infinite loop
INPUT
Your INPUT goes here!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QUESTIONS 10 / 10 completed
Queue
QUEUE
Write a C program to implement a Queue for managing online food delivery orders where orders are received and processed in the order they arrive. The program should allow adding new orders, processing completed orders, and displaying pending orders.
PROGRAM EDITOR
C
Run
Save
#include <stdio.h>
#define MAX 100
int queue[MAX];
int front = -1, rear = -1;
int isEmpty() {
return front == -1 && rear == -1;
}
int isFull() {
return rear == MAX - 1;
}
void enqueue(int value) {
if (isFull()) {
printf("Queue Overflow! Cannot add more orders.\n");
return;
}
OUTPUT

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QUESTIONS 10 / 10 completed

Queue Using Linke

QUEUE USING LINKED LISTS -2

Write a C program to implement a Queue using Linked Lists for managing network packets in a router. Packets should be transmitted in the order received, and the program must support dynamic packet insertion and removal. Sample Input: Insert Packets 11, 12, 13 Transmit

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>
#include <stdlib.h>

// Structure to represent a network packet
struct Packet {
    int packetID;
    struct Packet* next;
};

// Pointers to manage the queue
struct Packet* front = NULL;
struct Packet* rear = NULL;

// Function to insert a packet (Enqueue)
void insertPacket(int id) {
    struct Packet* newNode = (struct Packet*)malloc(sizeof(struct Packet));
    newNode->packetID = id;
    newNode->next = NULL;

    if (rear == NULL) {
        front = rear = newNode;
```

OUTPUT

```
Transmitted Packet: 11
```

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Nithi 19257

QUESTIONS 10 / 10 completed

Circular Queue

CIRCULAR QUEUE

Write a C program to implement a Circular Queue for a traffic signal management system where vehicles are processed lane by lane in a cyclic manner. The system should continuously rotate through available lanes while maintaining queue order. Sample Input: Vehicles 21, 22, 23

PROGRAM EDITOR

C

Run Save

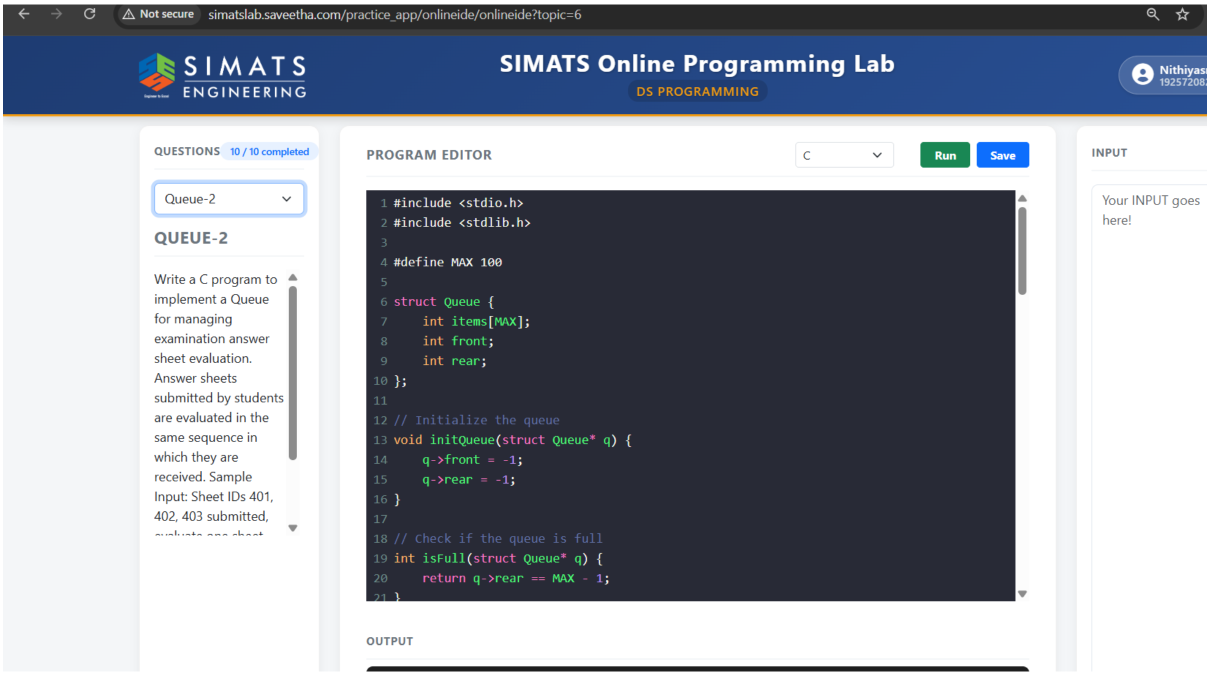
```c
#include <stdio.h>
#include <stdlib.h>

#define SIZE 5

int items[SIZE];
int front = -1, rear = -1;

// Check if the queue is full
int isFull() {
    if ((front == rear + 1) || (front == 0 && rear == SIZE - 1)) return 1;
    return 0;
}

// Check if the queue is empty
int isEmpty() {
    if (front == -1) return 1;
    return 0;
}

// Add an element
```

INPUT

Your INPUT go here!

OUTPUT

```
Vehicle Arrived: 21
```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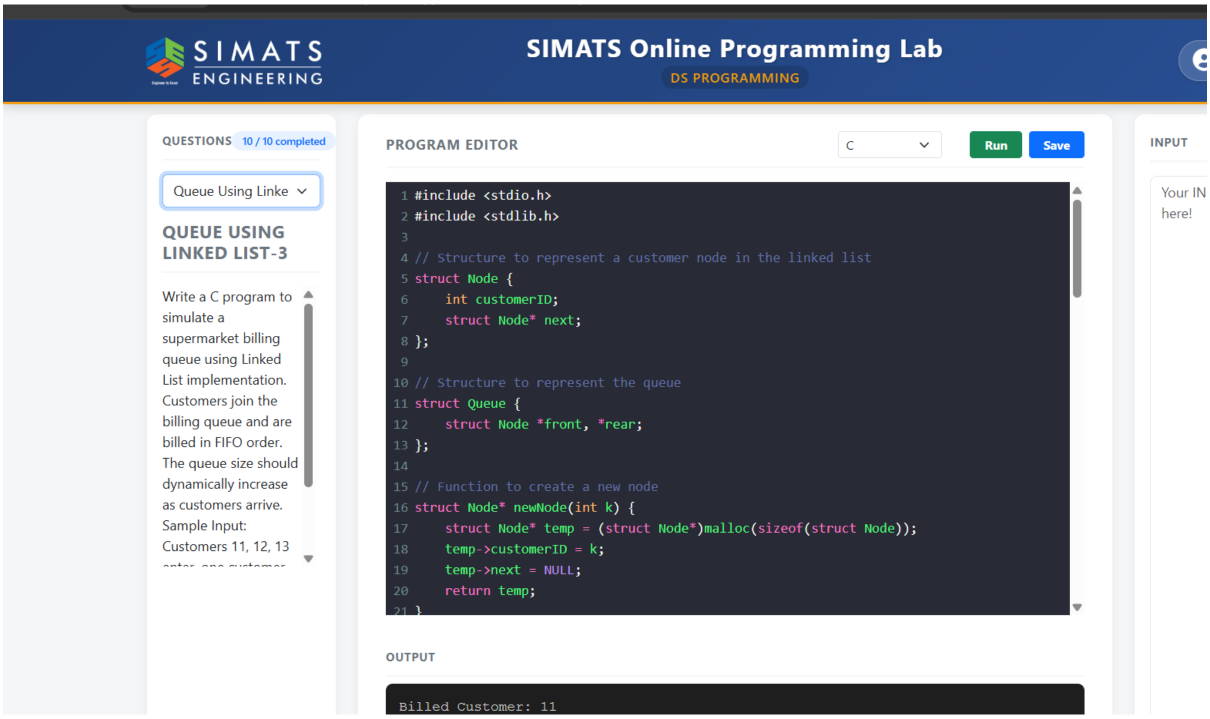
QUESTIONS 10 / 10 completed
Queue-2
QUEUE-2
Write a C program to implement a Queue for managing examination answer sheet evaluation. Answer sheets submitted by students are evaluated in the same sequence in which they are received. Sample Input: Sheet IDs 401, 402, 403 submitted,
PROGRAM EDITOR
C
Run
Save
#include <stdio.h>
#include <stdlib.h>
#define MAX 100
struct Queue {
int items[MAX];
int front;
int rear;
};
// Initialize the queue
void initQueue(struct Queue* q) {
q->front = -1;
q->rear = -1;
}
// Check if the queue is full
int isFull(struct Queue* q) {
return q->rear == MAX - 1;
OUTPUT
INPUT
Your INPUT goes here!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QUESTIONS 10 / 10 completed
Queue Using Linke
QUEUE USING LINKED LIST-3
Write a C program to simulate a supermarket billing queue using Linked List implementation. Customers join the billing queue and are billed in FIFO order. The queue size should dynamically increase as customers arrive. Sample Input: Customers 11, 12, 13
PROGRAM EDITOR
C
Run
Save
#include <stdio.h>
#include <stdlib.h>
// Structure to represent a customer node in the linked list
struct Node {
int customerID;
struct Node* next;
};
// Structure to represent the queue
struct Queue {
struct Node *front, *rear;
};
// Function to create a new node
struct Node* newNode(int k) {
struct Node* temp = (struct Node*)malloc(sizeof(struct Node));
temp->customerID = k;
temp->next = NULL;
return temp;
OUTPUT
Billed Customer: 11
INPUT
Your IN
here!